RESEARCH METHODS FOR COMPUTING AND TECHNOLOGY (CT098-3-2-RMCT)

INDIVIDUAL ASSIGNMENT

PART 2 - Methodology

Name	**Low Yi Bo**
TP Number	**TP076096**
Lecturer	**Dr. Hemalatha A/P Ramalingam**
Intake	**APD2F2506CS(DA)**
Date Assigned	**13th Jan 2026**
Submission Date	**16th Mac 2026**

Limitations of Traditional ERP Demand Forecasting in Retail SMEs: An Interpretable TKAN-Based Approach

Low Yi Bo (TP076096)

TP076096@mail.apu.edu.my

Abstract - Although retail Small and Medium Enterprises (SMEs) play a vital role in driving the Malaysian economy, they often suffer extreme operational disruptions owing to legacy Enterprise Resource Planning (ERP) forecasting tools. Within the highly fluid e-commerce sector, traditional linear models do not take into account the non linear trends of consumer purchasing behaviour, leading to a repetition of prediction mistakes, and the costly 'Bullwhip Effect.' This research fills this gap in technology development by presenting a Temporal Kolmogorov-Arnold Network (TKAN) architecture. Departing from the 'black box' nature of traditional neural networks, TKANs implement learnable B-spline activation functions to implement at the edges of the networks instead of fixed nodes. This has the dual benefit of capturing non-linear market dependencies, which are often what is most complex, and providing at the same time symbolic interpretability. By applying this transparent layer of intelligence to existing ERP environments, the proposed solution seeks to dramatically reduce errors in forecasts, thereby reducing inventory holding costs and to give managers of SMEs the mathematically verifiable precision required for them to be able to trust automated planning in a digital economy.

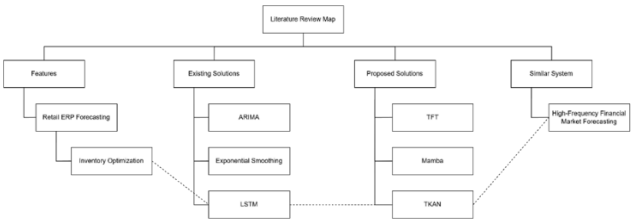
1.0 Introduction

Small and medium enterprises (SMEs) are a vital part of the Malaysian retail and e-commerce supply chains and account for 38.4% national GDP as of 2024. Effective inventory management is critical to these businesses as it directly affects operational costs, cash flow and financial stability. As competition in the market rises and profit margins tend to reduce, SMEs need to maximise the inventory turnover with minimum holding costs. However, their ability to keep inventories planned efficiently with economical methods is challenged increasingly due to higher market volatility and unpredictable demand routes. Traditional forecasting methods such as moving averages, exponential smoothing, and ARIMA models are designed to model linear and steady trends, making them less effective in greatly dealing with complex and non-linear demand fluctuations in contemporary and fashion retail markets. Although Malaysian SMEs have been using digital solutions like ERP systems to address business issues, many ERP forecasting modules still use static statistical models that find it difficult to adapt to rapidly changing market conditions.

Forecasting inaccuracies in these types of environments can flow through the supply chain and result in the Bullwhip Effect - In which small variations in demand cause large variations in inventory and orders. To overcome these limitations, there have been advanced machine learning models to represent the non-linear demand behaviors. While deep learning models such as LSTM are able to learn long-term temporal dependencies, these models are not easy to interpret which makes it challenging for small and mid-size enterprise (SME) managers to put their faith in and adopt them. Recent developments like the Temporal Kolmogorov-Arnold Network (TKAN), however, promise to be an alternative approach that replaces fixed and linear weights with learnable B-spline activation functions to allow

for both accurate non-linear forecasting and mathematical interpretability. Therefore, in an effort to replace the legacy ERP forecasting modules with an interpretive TKAN-based forecasting model, this research aims to improve demand prediction accuracy, and to reduce the supply chain disruptions caused by the Bullwhip Effect, and to facilitate more reliable inventory planning of Malaysian SMEs.

2.0 Literature Review



2.1 Features

Enterprise Resource Planning (ERP) system serves as the main operational backbone for management of business resources and supply chain stability. Within these infrastructures, demand forecasting serves as the fundamental predictive machine that is meant to work with historic sales information and project future consumer demand. Despite their importance, traditional forecasting modules often rely on hard statistical assumptions that do not work with highly non-linear and erratic purchasing patterns typical of modern-day e-commerce markets (Chowdhury et al., 2025). Recent comparative analyses confirm the idea that univariate statistical techniques are far behind modern multivariate deep neural networks in terms of their ability to predict complex retail sales (Theodoridis & Tsadiras 2025). Consequently, abrupt changes in the market are often the cause of delayed system response times and lower predictive accuracy. Such functional shortcomings present a direct challenge in the process of supply chain operations and the consistency of supply chain planning for inventory distributed to vulnerable Malaysia SMEs.

Beyond these functional constraints, conventional predictive models have the problem of being unable to improvise sudden external shocks in the market. A critical vulnerability is found in their rigidity regarding dynamic shifts within the environment like viral social media trends or sudden promotional shows. Saricioglu, Genevois and Cedolin (2024) point out that the heaviness of reliance on these traditional, one-step forecasting methods have the effect of systematically exaggerating demand distortion, outing forth the Bullwhip Effect (Lee et al, 1997). Modern retail inventory optimization requires highly adaptive predictive frameworks that can react to rapidly changing consumer behaviors and high intensity localized demand spikes. If the forecasting mechanisms are based solely on the fact that they map the historical data linearly, their accuracy becomes extremely poor during disruptions in the market, thereby putting the financial health of the business on fundamentally invalid footing.

2.2 Existing Solutions

The viability of current demand forecasting systems is largely evaluated across three important criteria: predictive power amid non-linear market volatility, computational performance for legacy ERP integration and algorithmic transparency that is needed for managerial confidence. Traditional statistical models such as Autoregressive Integrated Moving Average (ARIMA) and Exponential Smoothing are computationally

light but mostly do not pass the accuracy hurdle. To try and fill this accuracy gap, modern supply chain systems have focused increasingly on the use of advanced machine learning models. Studies show that advanced forecasting using a multi-step machine learning algorithm outperforms traditional moving average by a large margin in terms of constraining the amplification of demand and improving order fulfillment rate (Sarıcıoğlu et al., 2024).

Among these solutions, standard Long Short-Term Memory (LSTM) networks have become a dominant baseline as they are superior to processes that can deal with sequential time-series data. However, although the standard LSTMs models are successful in fulfilling criteria of computational efficiency and general accuracy, the basic architecture of LSTMs has no explicit interpretability. Because their outputs are based entirely on hidden neural correlations, not transparent, mathematical rationales, these "black box" models are not good at providing actionable, verifiable insights to human decision-makers. As a consequence of this, SME managers cannot believe in the reasonings behind automated inventory recommendations and they systematically trigger manual managerial overrides, perpetuating the Bullwhip Effect.

2.3 Proposed Solutions

In addressing the shortcomings of the above standard black box models, it is a work to adapt the Temporal Kolmogorov-Arnold Network (TKAN) framework for legacy ERP systems. Introduced by Genet and Inzirillo (2024), the TKAN architecture can be considered as a critical paradigm shift, because it is the learnable B-spline activation functions applied to the edges of the network, which brings the management of temporal sequences directly into the heart of the T-KAN neural network. Unlike pure deep learning models, the proposed TKAN architecture makes use of symbolic regression that mathematically models the system's internal processing into readable and easily-understood formulas.

In order to understand the functional superiority of TKANs, it is necessary to explore the mathematical theorem on which its architecture has been built. Standard Multi-Layered Perceptrons (MLPs) and LSTMs are based upon fixed activation functions such as ReLU, Sigmoid or Tanh that are applied at the structural nodes. In contrast, the architecture of TKAN is based on the so-called Kolmogorov-Arnold representation theorem and can be represented by: (Liu et al., 2024). Instead of using fixed node activations, TKANs use univariate learnable B-spline functions on the edges of the network. As the network learns on the volatility of any retail data the training, these B-splines dynamically change their shapes to map the exact mathematical contours of the demand curve. This localised learning capability explicitly returns the learned demand function in the form of a symbolic equation (Genet & Inzirillo, 2024). Consequently, this enables the system to condense a complex temporal supply chain patterns into reachable, human readable algebraic expressions.

This spline-based interpretable machine learning has been very effective in the volatile forecasting domains. For example, some of the recent implementations of KAN principles in energy demand forecasting have been able to capture complex sequential relationships that traditional

models have been unable to detect (Zhang et al., 2025). Furthermore KAN architectures are more accurate in prediction than standard Multi-Layer Perceptrons (MLPs) with fewer numbers of trained parameters, proving the computational efficiency needed to be incorporated into SME legacy systems (Arslan & Dönmez, 2025). By pairing data-driven outputs with symbolic mathematical proofs, TKAN model enables human operators to validate the true non-linear market mechanisms of a demand spike to ensure the credibility of automation organically managing inventory planning during a crucial supply chain disruption.

2.4 Similar Systems

A recent study by Makinde (2026) titled "Temporal Kolmogorov-Arnold Networks (T-KAN) for High-Frequency Limit Order Book Forecasting: Efficiency, Interpretability, and Alpha Decay" has demonstrated the use of spline-based mathematical interpretability in deep learning models to forecast complex quantitative finance models. The architecture of their proposed system combined learnable B-spline activation functions and a temporal network to process real-time stock market trading data in an efficient way. By gathering high frequency limit order data during periods of high volatility in the markets, the researcher was able to allow the neural network to map exact, non-linear mathematical relationships during its training phase. As a result, the spline-based framework greatly enhanced the accuracy of real-time predictions of price movements and ensured that the outputs were mathematically clear, rather than opaque statistical correlations as usual baseline models would be.

Although Makinde (2026) applied this TKAN architecture in the quantitative finance industry in order to keep track of high-frequency trading processes, the current application differs from previous uses by applying this same technological foundation to the stabilization of retail ERP demand forecasting. The fact that the TKAN network's power to preserve strictly predictive and interpretable qualities is not diminished by pastoring severe challenges such as quickly shifting market shocks what makes the implications of the functional implementation strength during each request. By mathematically translating data-driven outputs based on symbolic regression formulas, the technique has managed to avoid the 'black box' effects of trust failures that are commonly displayed when using standard deep learning models. This strong performance across industries validates the integration of TKAN principles as a very viable and practical way to come up with trustworthy actionable inventory forecasts.

3.0 Methodology

This research takes a quantitative, experimental research methodology. It involves extraction, processing and statistical assessment of numerical time series data aimed at first benchmarking the computational efficiency of a new architecture of a neural network against the efficiency of the traditional forecasting models.

3.1 Target User

In this research, the different target users are classified according to the level of interaction with the forecasting system and their position in the retail supply chain. This classification figures out the difference between those who perform

the work of means of the software to make data-driven decisions and those who are the larger ecosystem of those who benefit from the stability of the operations.

3.1.1 Direct Users

The direct users of the proposed TKAN based forecasting system mainly include SME managers, inventory planners and procurement officers who are responsible for daily supply chain operations. These users interact directly with the ERP interface, feeding the historical sales data and promotional calendars and local market variables the model needs to operate. Unlike traditional "black box" neural networks, these users in the TKAN architecture are given symbolic mathematical interpretations of the predicted demand and, therefore, can check the logic behind making a forecast before they actually make a purchase order. By capitalizing upon this transparent intelligence, the direct users will be able to greatly lessen the amount of manual managerial overrides, which are usually the main factor driving the Bullwhip Effect, shifting their workflow from reactive management of a crisis to proactive, evidence-based inventory planning.

3.1.2 Indirect Users

Indirect users refer to the stakeholders who do not use the forecasting platform but are positive strong users of the results of its application. This category includes upstream suppliers and wholesalers, who benefit from the improved stability of orders, as well as more predictable demand patterns, that occur when SMEs use more precise forecasting tools. At the consumer level, retail customers are indirect beneficiaries who benefit from an improved product availability and fewer incidents of out-of-stock conditions for basic products. Furthermore, the SME business owners, institutional stakeholders are considered as indirect users as they benefit from the long-term financial health of the enterprise. By ensuring more efficient turnover of inventory and less holding costs, the system spreads better cash flow to these stakeholders and helps mathematicians verify its users with a verifiable tool to keep business going against the risk of volatile changes in a market.

3.2 Sampling Method

A mixed sampling strategy that combines elements of both purposive and cluster sampling methods is implemented for this research study in order to make the information amended technically viable for the training of the Temporal Kolmogorov-Arnold Network (TKAN) while being representative of the wider retail landscape. This approach is consistent with preexisting research frameworks where a tradeoff between specific issues relating to data quality and generalizable findings across a spectrum of population segments is required.

The first stage of the process involves the use of purposive sampling, which is a non-probability sampling method in which samples are chosen based on certain attributes that are relevant for the research objectives. For this study, the selection criteria are targeted at the retail SMEs in Malaysia with the established digital ERP infrastructure and at least twenty-four months of structured

historical sales records. This targeted selection is of critical importance since the mathematical transparency and predictive merit of the TKAN model is dependent on the quality of the time series data collected. By choosing SMEs that have a certain level of technological maturity, the research guarantees that more basic data is robust enough to support the learnable B-spline activation functions and symbolic regression that will be the requirement of the proposed architecture.

Second stage applies cluster sampling which is a probability-based technique in which the population is broken down into a number of naturally-occurring groups or clusters. In the context of this research, clusters are defined by sub-sectors of retailing, including fast-moving consumer goods (FMCG), electronics and apparel. Rather than looking at individual product SKUs alone, whole sub-sector data clusters are analyzed, to pick-up the unique and non-linear fluctuations in demand and seasonal spikes that exist for different retail niches. This approach aids in making the findings more representative as the model is verified against a range of market behaviours. Furthermore, there is a rational basis for mathematically evaluating the effectiveness of the system in terms of mitigating the Bullwhip Effect at the wider level of the retail supply chain by using cluster sampling which incurs lower administrative complexity.

3.3 Data Collection Method

The data collection process for this research is focused around the retrieval of high resolution, longitudinal, secondary data, from the internal Enterprise Resource Planning (ERP) systems of the retail SMEs participating in the research. This method is chosen in order to guarantee the integrity and granularity of the temporal datasets that are needed to train and to validate the TKAN-based forecasting architecture. By mining data directly from the digital record, the study removes the subjectivity of bias associated with manual reporting and use a mathematical foundation for the objective analysis of non-linear consumer behaviors.

The main source of the data is historical sales transactions and inventory data for at least two years. Executing this data collection requires certain technical resources like access to the ERP database exports of the SMEs in question like a SQL dump or a CSV file and computational software environments such as Python, PyTorch or TensorFlow to process the inputs of the data. Specific data variables to be extracted include daily unit sales, stock positions of SKUs, lead times for stock replenishment and history of promotional markers. Additionally, external market indicators, such as regional economic indicators, or viral product metadata, may also be incorporated into the information gathering process to act as a supplemental input to the symbolic regression layer of the model.

From the side of technical implementation, the development and training part of the TKAN architecture will be carried out through an up-to-date machine learning ecosystem based on Python. The core neural network will be

built with the PyTorch open-source machine learning framework (Paszke et al 2019), by integrating some of the specialized libraries of PyKAN that will enable the deployment of Kolmogorov-Arnold mathematical layers. One of the great benefits of choosing this particular architecture based on a spline is one of computational efficiency. The phases in the compilation, training and testing of this research including can be viably executed on standard hardware accelerated by consumer grade Graphics Processing Units (GPUs). This hardware efficiency is indeed a reflection of the constrained IT budgets and legacy infrastructural realities of the type of Malaysian retail SMEs that this research is trying to assist.

Following data collection the results of data collection will be interpreted with a rigorous statistical analysis. The accuracy of the predictive model using TKAN will be quantitatively assessed using traditional industry metrics, main inclusion Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE). Choosing the right statistical evaluation metrics is imperative for actualizing the value of neural network performance into supply chain value. The predictive accuracy of the TKAN model will be mostly evaluated through the use of Mean Absolute Percentage Error (MAPE), Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), which are standard benchmarks of time series forecasting reliability (Hyndman & Koehler, 2006). MAPE is chosen as it forecasts errors as a percentage which is an intuitive measure for SME managers. However, since MAPE may be skewed when actual demand declines to zero, RMSE and MAE will be used as a secondary measure. RMSE is especially important in cases of supply chain stabilization because it dollars Chance errors mathematically more heavily than small errors. By optimizing the TKAN architecture towards the minimization of RMSE, the system has an explicit emphasis on the avoidance of a high magnitude of prediction failures that create the Bullwhip Effect.

To counter the ethical and security concerns, mainly regarding data privacy and corporate confidentiality, a rigid de-identification procedure is adopted. In compliance with data protection measures like Malaysia's Personal Data Protection Act (PDPA) data retrieval is done focusing on the volume and time stamp of numerical transaction data alone, completely eliminating any sensitive customer data or proprietary vendor identities. The raw data will then go through a rigorous preprocessing phase involving the inclusion of normalization and dealing with missing values which would then be fed to the input layers of the neural network.

Because of the inherent nature of raw time-series data extracted from legacy SME environments, that data is notoriously susceptible to noise, manual entry errors and systemic inconsistencies, a thorough preprocessing pipeline is a mandatory part of this methodology. First, missing points that were due to system downtime or to unrecorded sales on the weekends will be treated with forward-fill interpolation to preserve chronological integrity of the time series.

Second, to deal with the extreme anomalies such as data entry typos which might accidentally trick the model to expect non-existing demand spikes, Isolation Forest algorithm or Z-score thresholding will be used which will identify and smooth extreme outliers without on a real promotional peaks.

Furthermore, since neural networks need to have standardised inputs to graduate descent convergence in an efficient manner, all numerical sales volumes and inventory levels will be Min-Max normalised, which requires the data to be strictly scaled between a range of 0 and 1. Finally, to provide the model with a predictive validity and avoid the challenge of data leaking, the continuous 24 months time frame will be split chronologically instead of being divided by a random split. A standard temporal split will be enforced with the first 70% (around 17 months) of the timeline as the training set, the next 15% as the validation set to tune hyperparameters, and finally the last 15% will be withheld as the blind testing set to simulate on seen data (i.e. the real world market through unseen forecasting).

One of the main practical limitations of this methodology is that it is strongly dependent on the quality of the data that is being stored in the underlying SME ERP systems. Inconsistent historical logging, missing values or unstructured data entries by SME staff could have a seriously adverse effect in the neural network training phase. However, the aforementioned rigorous preprocessing and data normalization techniques would be used to control for these confounding variables and reduce possible error, and ensure that the resultant data set would be a reliable reflection of the volatile market dynamics that Malaysian retail SMEs currently struggle with.

4.0 Conclusion

This research methodology provides a strong framework in solving the critical inefficiencies of traditional ERP demand forecasting in Malaysian retail SME sector. By moving from fixed, linear statistical models towards a Temporal Kolmogorov-Arnold Network (TKAN) architecture, the work offers a channel towards a high precision predictive accuracy while also giving fundamental mathematical interpretability. The methodology guarantees that the solution proposed is based on operational data obtained in the real world through a systematic sampling and collection process, focusing on those digitally-ready SMEs where the impact of the Bullwhip Effect is strongest.

Ultimately, the combination of learnable B-spline activation functions and the use of symbolic regression is meant to overcome an unfortunate “trust gap” that presently limits the adoption of deep learning in supply chain management. By offering SME managers mathematically verifiable inputs instead of opaque “black box” outputs, not only is this approach reducing inventory holding costs and operational disruptions, it is also enabling decision-makers to have confidence in their decision-making through the autonomy of automated intelligence. The success of this research will prove that high-tech, yet practical AI architectures can be developed that provide a scalable model for digital resilience in a volatile e-commerce economy.

5.0 References

Arslan, E., & Dönmez, A. (2025). Natural gas consumption forecasting with

Kolmogorov–Arnold Networks: A comparison with MLP. Sakarya University Journal of Computer and Information Sciences. http://saucis.sakarya.edu.tr/en/pub/article/1759966

Chowdhury, S., et al. (2025). The impact of AI-based demand sensing on inventory optimisation and bullwhip effect mitigation. Journal of Marketing & Social Research.

Genet, R., & Inzirillo, C. (2024). TKAN: Temporal Kolmogorov-Arnold Networks. arXiv preprint arXiv:2405.07344. https://arxiv.org/abs/2405.07344

Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. International Journal of Forecasting, 22(4), 679-688.

Lee, H. L., Padmanabhan, V., & Whang, S. (1997). Information distortion in a supply chain: The bullwhip effect. Management Science, 43(4), 546-558.

Liu, Z., Wang, Y., Vaidya, S., Ruehle, F., Halverson, J., Soljačić, M., ... & Tegmark, M. (2024). KAN: Kolmogorov-Arnold Networks. arXiv preprint arXiv:2404.19756.

Makinde, O. (2026). Temporal Kolmogorov-Arnold Networks (T-KAN) for high-frequency limit order book forecasting: Efficiency, interpretability, and alpha decay. Quantitative Finance Studies.

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., ... & Chintala, S. (2019). PyTorch: An imperative style, high-performance deep learning library. Advances in Neural Information Processing Systems, 32.

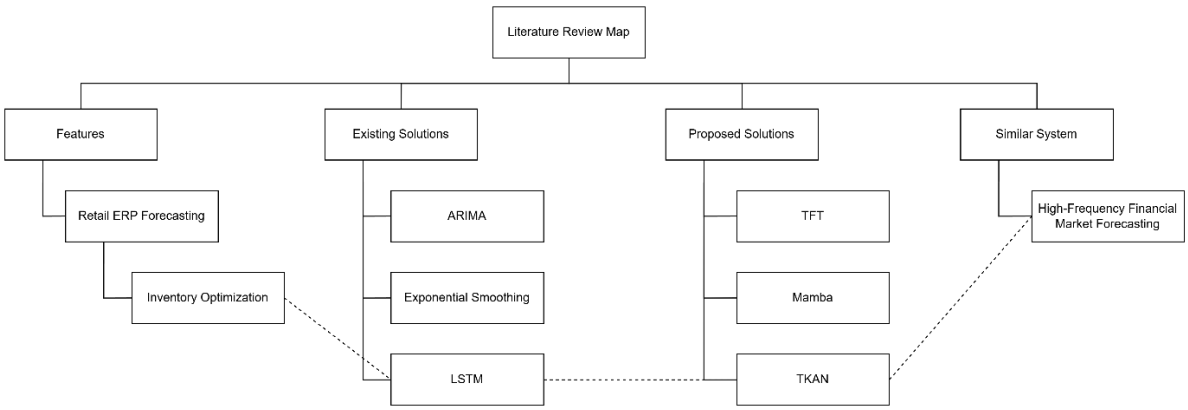
Sarıcıoğlu, M., Genevois, M. E., & Cedolin, B. (2024). Analyzing one-step and multi-step forecasting to mitigate the bullwhip effect and improve supply chain performance. IEEE Access. https://ieeexplore.ieee.org/document/10772207/

Theodoridis, I., & Tsadiras, A. (2025). Retail demand forecasting: A comparative analysis of deep neural networks and the proposal of LSTMixer, a linear model extension. Information, 16(7), 596. https://doi.org/10.3390/info16070596

Zhang, X., Cui, Y., & Yan, Z. (2025). Integrating Kolmogorov–Arnold Networks with time series prediction framework in electricity demand forecasting. Energies, 18(6), 1365. https://doi.org/10.3390/en18061365

6.0 Appendix

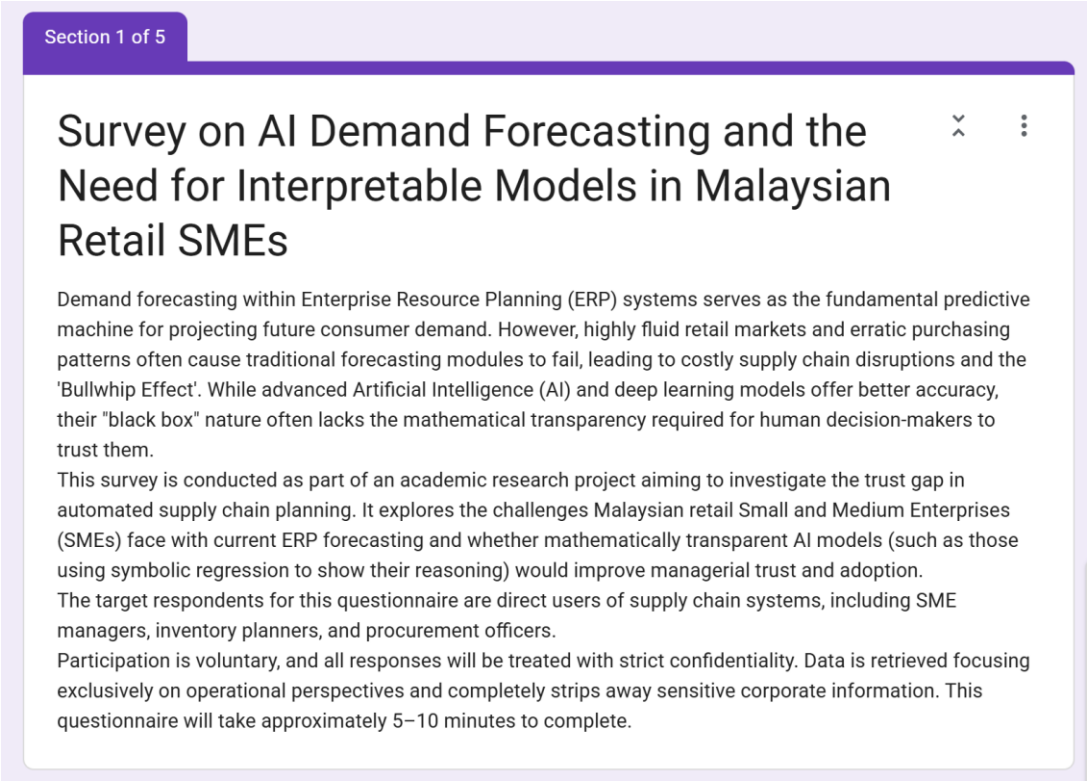
1. Literature Review Map



2. Similar System

Title	Author	Year	Technology Used	Sampling Method	Result	Limitations	Industry
T-KAN for High-Frequency Limit Order Book Forecasting	O. Makinde	2026	TKAN (B-spline activation)	High-frequency market data extraction	High accuracy & transparency over opaque models	Untested on retail/supply chain data	Finance

3. Sample Survey Questions



Section 1 of 5

Survey on AI Demand Forecasting and the Need for Interpretable Models in Malaysian Retail SMEs

Demand forecasting within Enterprise Resource Planning (ERP) systems serves as the fundamental predictive machine for projecting future consumer demand. However, highly fluid retail markets and erratic purchasing patterns often cause traditional forecasting modules to fail, leading to costly supply chain disruptions and the 'Bullwhip Effect'. While advanced Artificial Intelligence (AI) and deep learning models offer better accuracy, their "black box" nature often lacks the mathematical transparency required for human decision-makers to trust them.

This survey is conducted as part of an academic research project aiming to investigate the trust gap in automated supply chain planning. It explores the challenges Malaysian retail Small and Medium Enterprises (SMEs) face with current ERP forecasting and whether mathematically transparent AI models (such as those using symbolic regression to show their reasoning) would improve managerial trust and adoption.

The target respondents for this questionnaire are direct users of supply chain systems, including SME managers, inventory planners, and procurement officers.

Participation is voluntary, and all responses will be treated with strict confidentiality. Data is retrieved focusing exclusively on operational perspectives and completely strips away sensitive corporate information. This questionnaire will take approximately 5–10 minutes to complete.

Section 2 of 5

Professional Background & ERP Usage

This section aims to understand your role within the retail supply chain and your familiarity with inventory management systems.

Which of the following best describes your current role? *

- SME Owner / General Manager
- Inventory Planner / Controller
- Procurement / Supply Chain Officer
- IT / ERP System Administrator
- Other: _____

How many years of experience do you have in retail inventory management or supply chain operations? *

- Less than 1 year
- 1 - 3 years
- 4 - 7 years
- More than 7 years

What type of forecasting methods does your current ERP or inventory system primarily rely on? *

- Basic statistical models (e.g., Moving Averages, Exponential Smoothing, ARIMA)
- Advanced Machine Learning / AI models
- Manual calculations (e.g., Excel spreadsheets)
- I am not sure

How often do you experience sudden, unpredictable spikes or drops in consumer demand (e.g., from viral social media fads or localized promotions)? *

- Rarely
- Occasionally
- Frequently
- Very Frequently

Section 3 of 5

Current Forecasting Challenges & The Bullwhip Effect

This section explores the limitations you might face with your current forecasting tools during volatile market conditions.

My current ERP forecasting tool struggles to adapt to rapidly changing, non-linear market conditions. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

When sudden market disruptions occur, the forecasting system often provides delayed or inaccurate recommendations. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

I frequently have to manually override the system's automated inventory recommendations because I do not trust its accuracy. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Small variations in consumer demand often cause exaggerated disruptions (the Bullwhip Effect) in our inventory levels and supplier orders. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Section 4 of 5

Trust & The AI "Black Box" Barrier

Advanced AI can improve accuracy, but many operate as "black boxes" where the logic behind the prediction is hidden. This section explores your trust in such systems.

I would be hesitant to trust an AI-generated purchase order if the system could not explain how it arrived at that specific number. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

The lack of transparency in how deep learning models generate recommendations makes them unsuitable for critical financial decisions. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

I prefer relying on traditional statistical methods (even if they are less accurate) because their logic is easier to understand than complex AI. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

My main concern about automated AI forecasting is that I cannot verify whether the recommendation actually accounts for specific local market variables. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Section 5 of 6

Attitude Toward Interpretable AI Solutions

Imagine a new AI architecture (like a Temporal Kolmogorov-Arnold Network) that compresses complex market data into human-readable mathematical formulas, allowing you to clearly see the reasoning behind a demand spike.

Providing transparent, verifiable mathematical formulas alongside an AI forecast would significantly increase my trust in the system. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

If I could read and verify the exact variables the AI used to predict a demand spike, I would be less likely to manually override the system. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Interpretable AI models would shift my workflow from reactive crisis management to proactive, confident inventory planning. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Overall, integrating mathematical transparency into ERP forecasting modules is essential for the long-term adoption of AI in SME supply chains. *

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree